Introducción al modelo OMOP-CDM

Sesión 4.- Publicación y explotación

Alberto Labarga – Febrero 2026

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Febrero
2026

OMOP-CDM
El curso

El curso tiene como objetivo proporcionar una comprensión profunda del modelo de datos **OMOP-CDM** de la OHDSI, incluida su **estructura**, tablas clave y **vocabularios**.

Abordaremos también como **transformar** datos al modelo y explorar y analizar la **calidad** de los datos transformados.

L	M	X	J	V	S	D
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	1
2	3					

Objetivos del curso

01 Introducción al modelo OMOP-CDM, a sus tablas y sus vocabularios

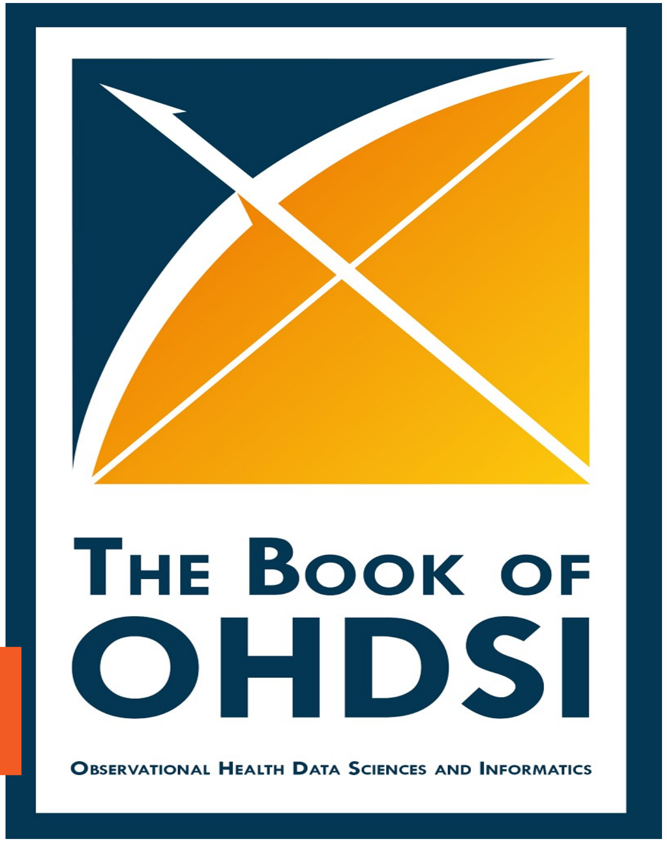
02 Transformación de datos al modelo OMOP-CDM utilizando la suite de herramientas de la OHDSI

03 Análisis de calidad y explotación de una base de datos OMOP-CDM

Recursos de aprendizaje

Sitio web: https://github.com/alabarga/curso-omop-cdm

Libro https://ohdsi.github.io/TheBookOfOhdsi/

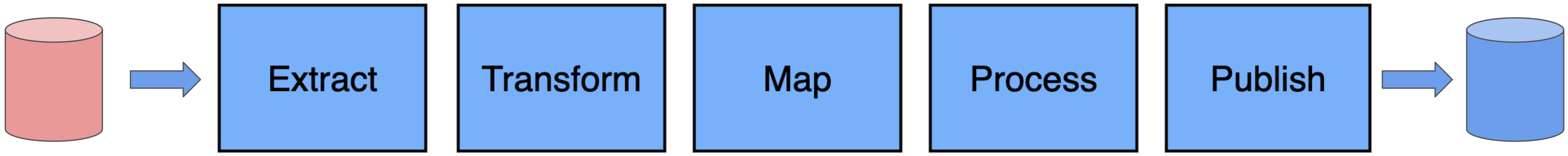


OMOP-CDM

Publicación y explotación de datos

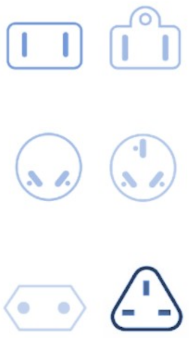

SJD
Sant Joan de Déu
Barcelona · Hospital
INGEMM
idis
IMIB
IMIBIC
IBSAL

TRANSFORMACIÓN
AL MODELO OMOP-CDM



TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish



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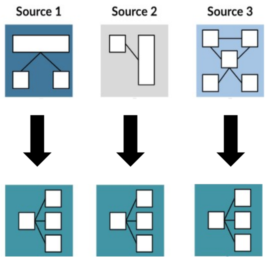

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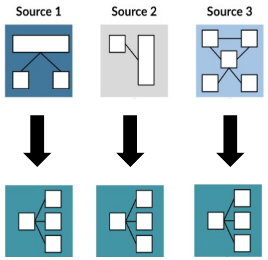


WHITE RABBIT

RABBIT IN A HAT

dbt

PROFILING

ydata-profiling

patients Profiling Report

Overview Variables Interactions Correlations Missing values Sample

Overview

Brought to you by YData

Overview Alerts 21 Reproduction

Dataset statistics

Number of variables	28
Number of observations	111
Missing cells	454
Missing cells (%)	14.6%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	24.4 KiB
Average record size in memory	225.2 B

Variable types

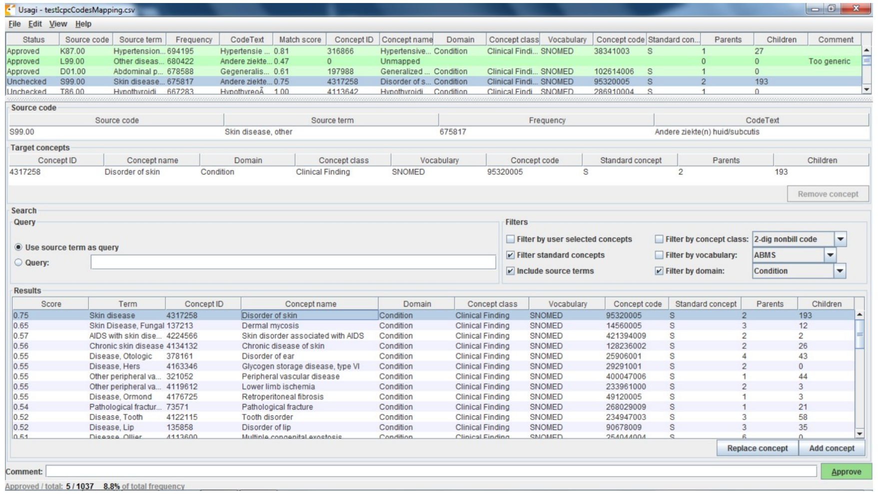
Text	21
Numeric	7

https://docs.profiling.ydata.ai/latest/

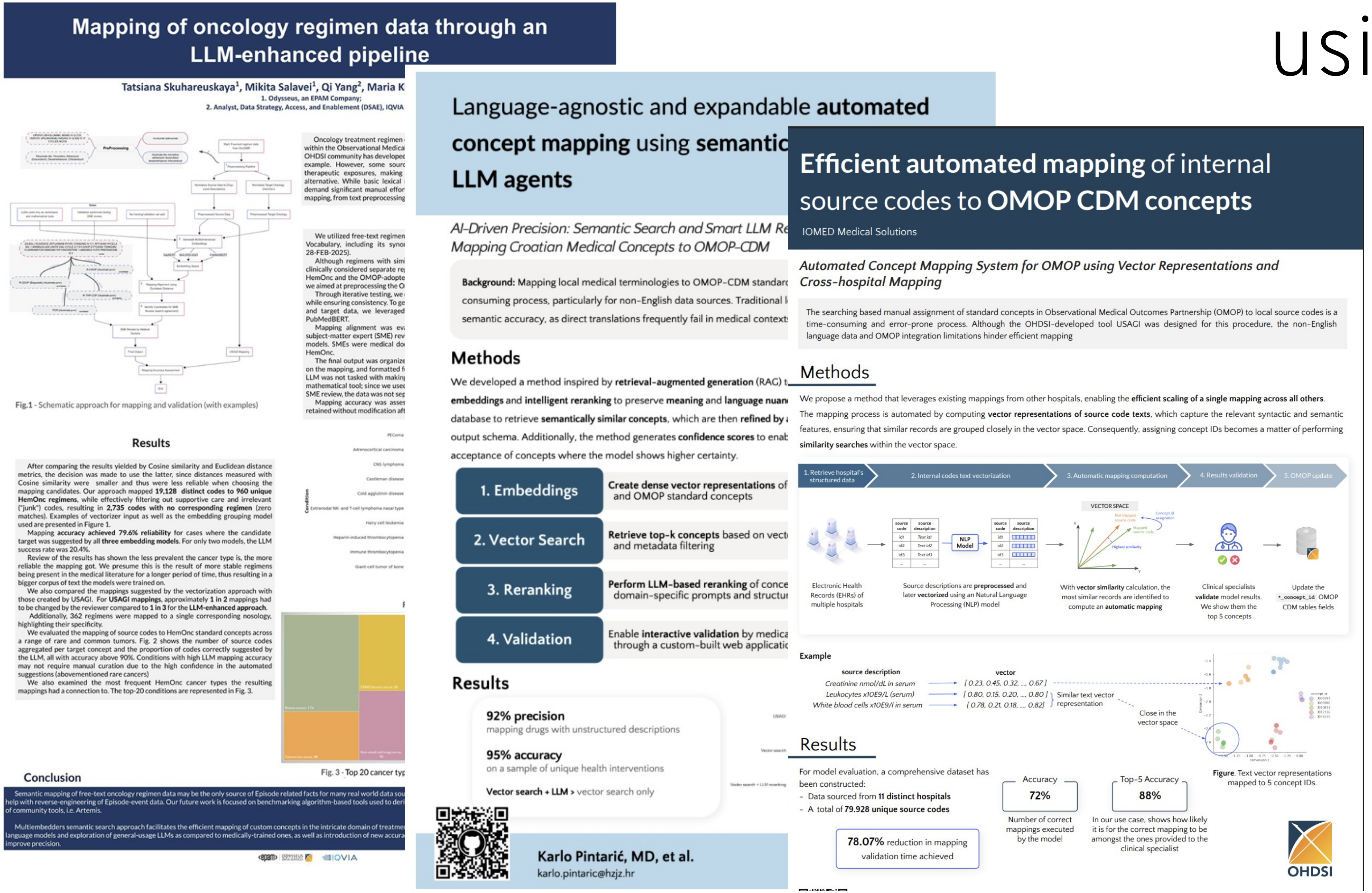
TRANSFORMACIÓN
AL MODELO OMOP-CDM

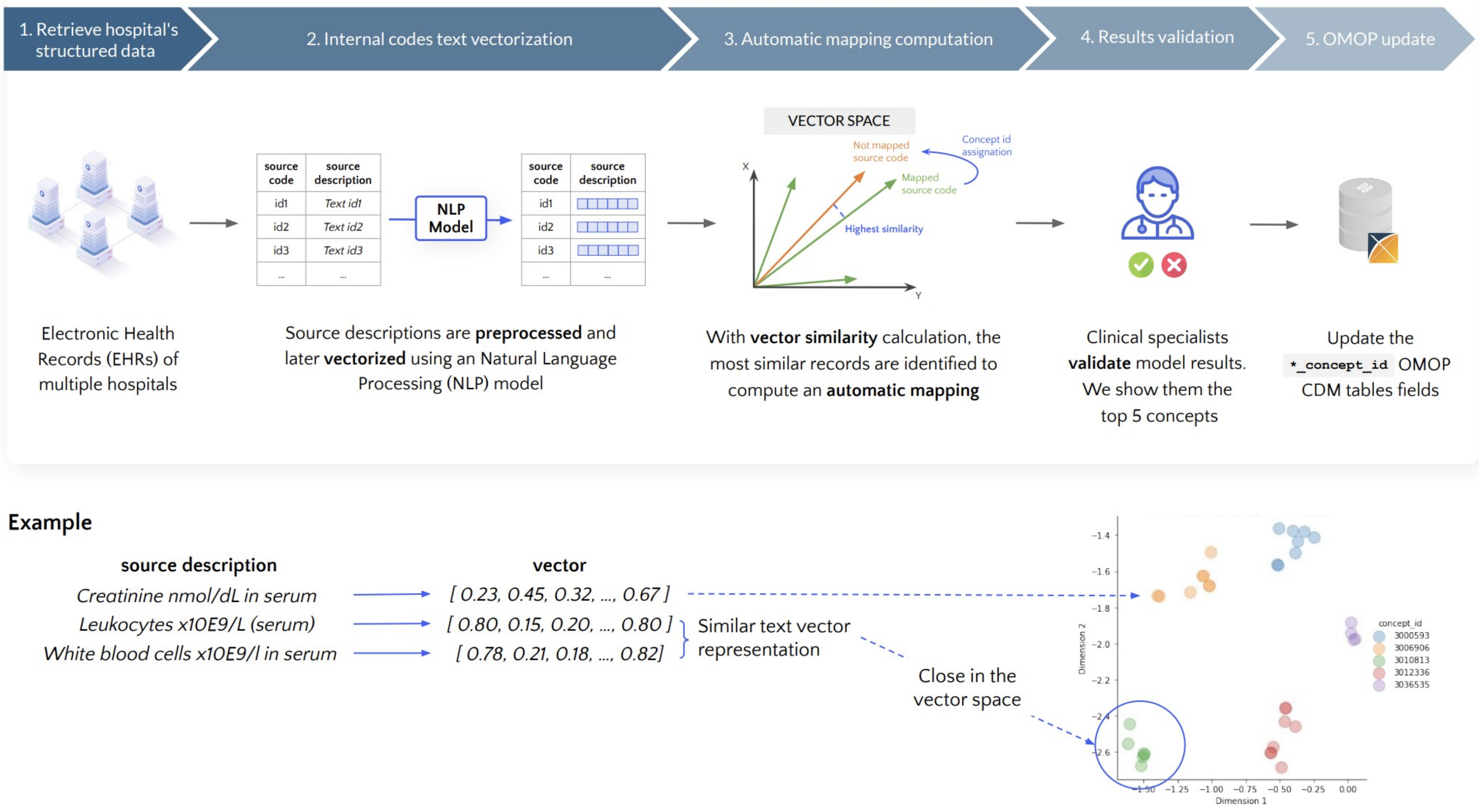
- Extract
- Transform
- Map
- Process
- Publish

USAGI



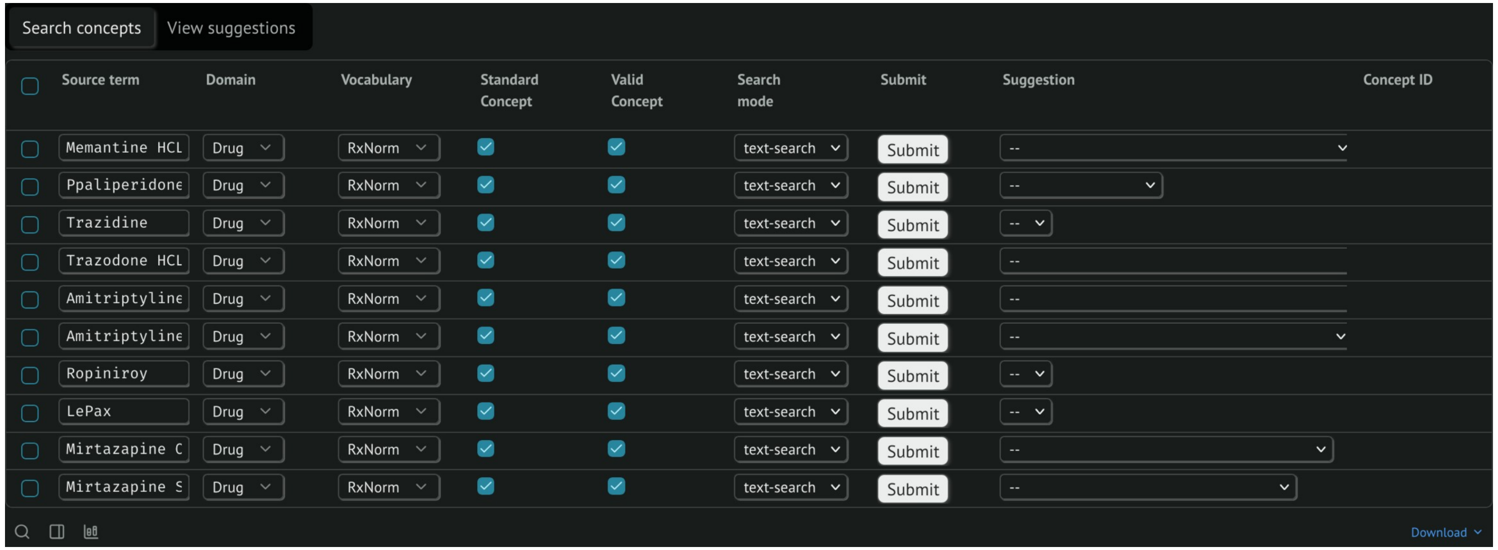
MAPPING
using LLMs





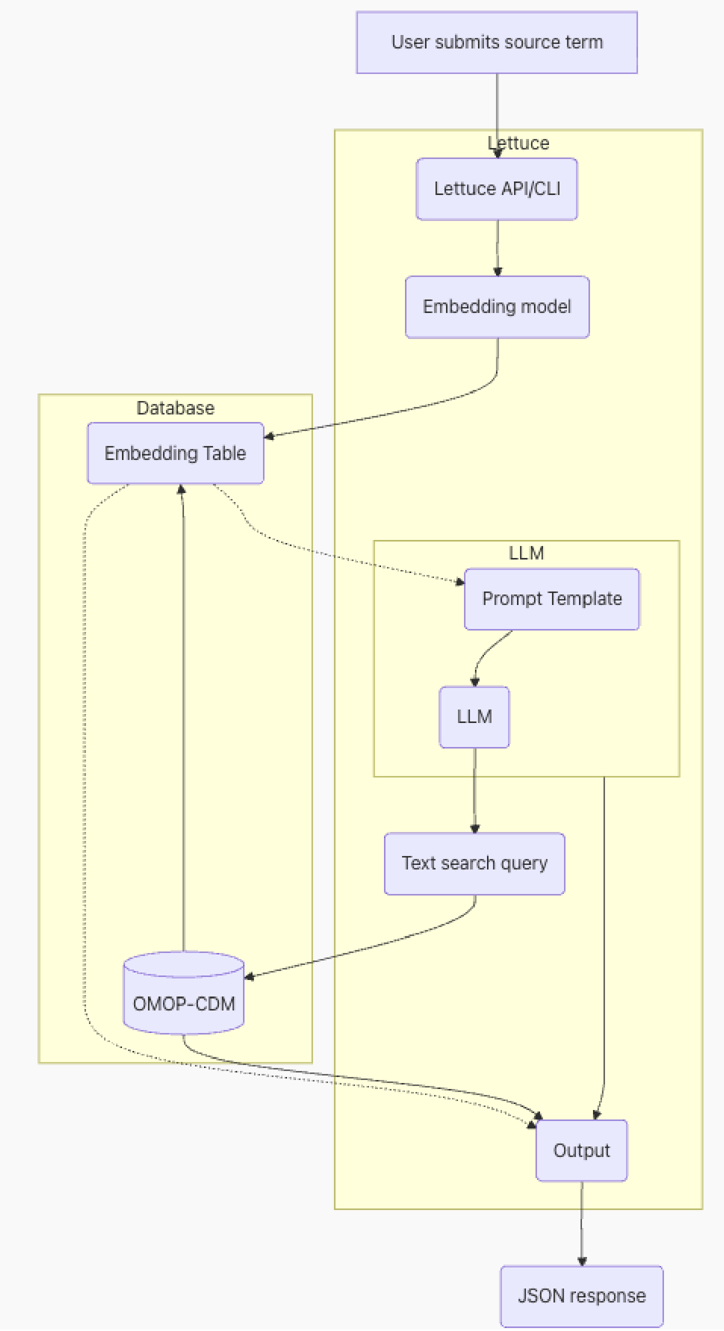
1. Retrieve hospital's structured data
2. Internal codes text vectorization
3. Automatic mapping computation
4. Results validation
5. OMOP update
source code
source description
id1
Text id1
id2
Text id2
id3
Text id3
NLP Model
VECTOR SPACE
Not mapped source code
Concept id assignation
Mapped source code
Highest similarity
Electronic Health Records (EHRs) of multiple hospitals
Source descriptions are preprocessed and later vectorized using an Natural Language Processing (NLP) model
With vector similarity calculation, the most similar records are identified to compute an automatic mapping
Clinical specialists validate model results. We show them the top 5 concepts
Update the *_concept_id OMOP CDM tables fields
Example
source description
vector
Creatinine nmol/dL in serum
Leukocytes x10E9/L (serum)
White blood cells x10E9/l in serum
[0.23, 0.45, 0.32, ..., 0.67]
[0.80, 0.15, 0.20, ..., 0.80]
[0.78, 0.21, 0.18, ..., 0.82]
Similar text vector representation
Close in the vector space
Dimension 1
Dimension 2
concept_id
3000593
3006906
3010813
3012336
3036535

Lettuce



Source term	Domain	Vocabulary	Standard Concept	Valid Concept	Search mode	Submit	Suggestion	Concept ID
Memantine HCL	Drug	RxNorm	✓	✓	text-search	Submit	--	
Ppaliperidone	Drug	RxNorm	✓	✓	text-search	Submit	--	
Trazidine	Drug	RxNorm	✓	✓	text-search	Submit	--	
Trazodone HCL	Drug	RxNorm	✓	✓	text-search	Submit	--	
Amitriptyline	Drug	RxNorm	✓	✓	text-search	Submit	--	
Amitriptyline	Drug	RxNorm	✓	✓	text-search	Submit	--	
Ropiniroy	Drug	RxNorm	✓	✓	text-search	Submit	--	
LePax	Drug	RxNorm	✓	✓	text-search	Submit	--	
Mirtazapine C	Drug	RxNorm	✓	✓	text-search	Submit	--	
Mirtazapine S	Drug	RxNorm	✓	✓	text-search	Submit	--	

https://health-informatics-uon.github.io/lettuce/



TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish

El paciente acude con su madre Pepita **PER**, con la que vive en Calle Córcega 23 **LOC**, refiriendo dolor abdominal.

El paciente acude con su madre Lucía **PER**, con la que vive en Calle Londres **LOC**, refiriendo dolor abdominal.

TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract → Transform → Map → Process → Publish

Acude con dolor abdominal derecho de 2 días de duración. Pauta de vacunación completa.

Finding *dolor abdominal.*

Spatial Concept *derecho.*

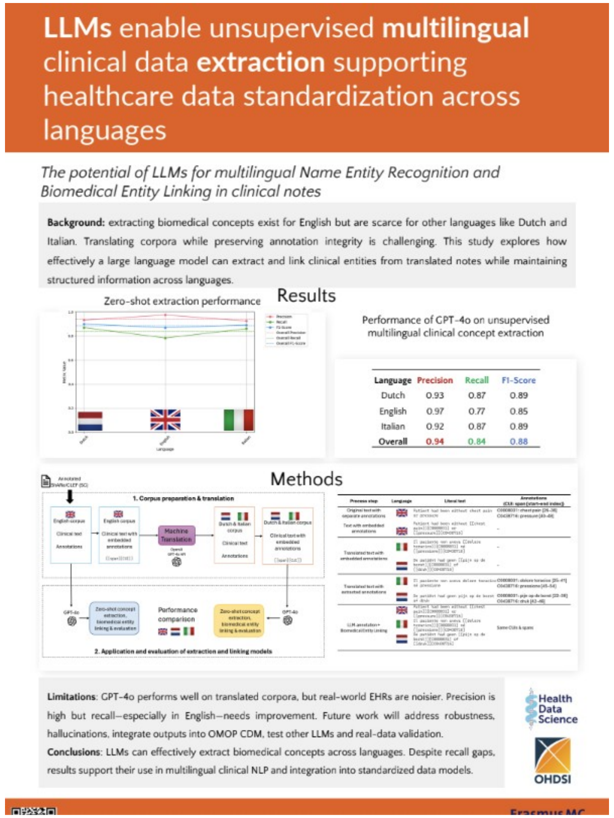
Temporal Concept *2 días, duración.*

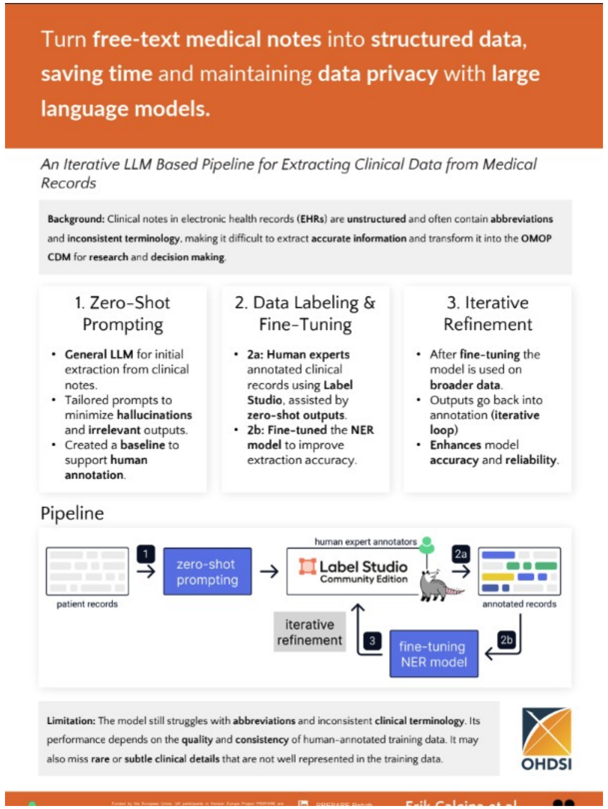
Therapeutic or Preventive Procedure *vacunación.*

Qualitative Concept *completa.*

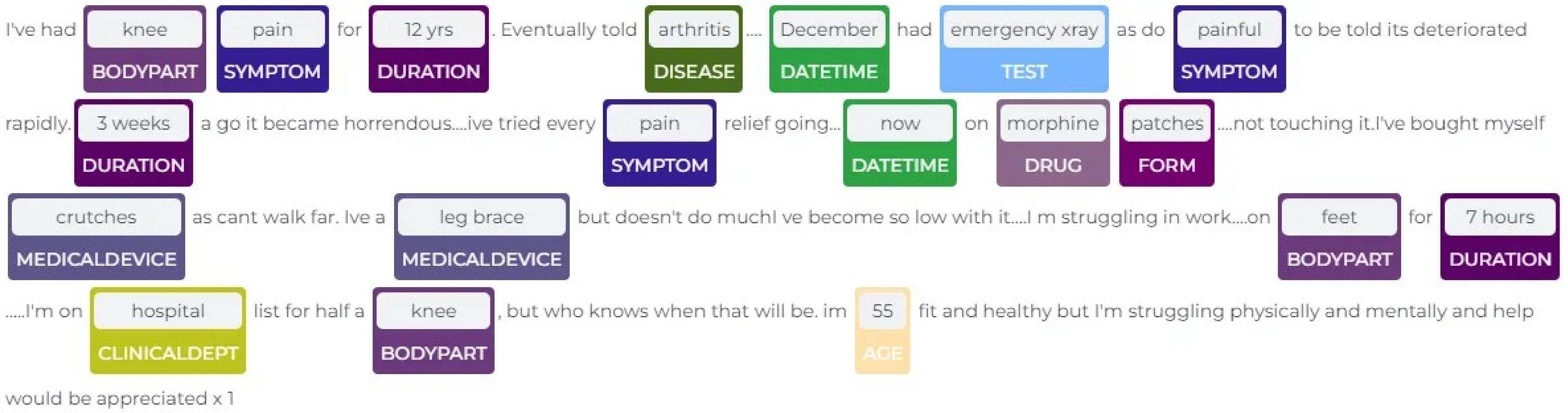
PROCESADO

DEL LENGUAJE NATURAL



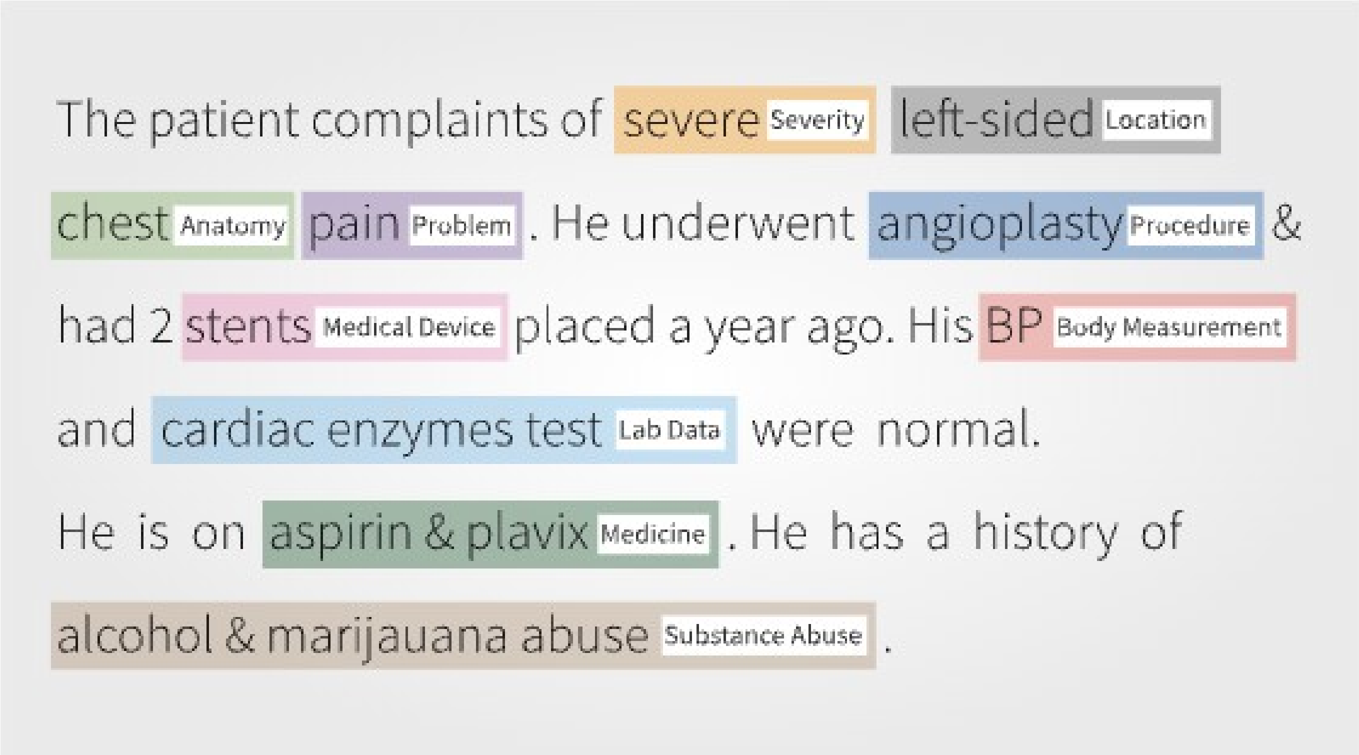


PROCESADO
DEL LENGUAJE NATURAL



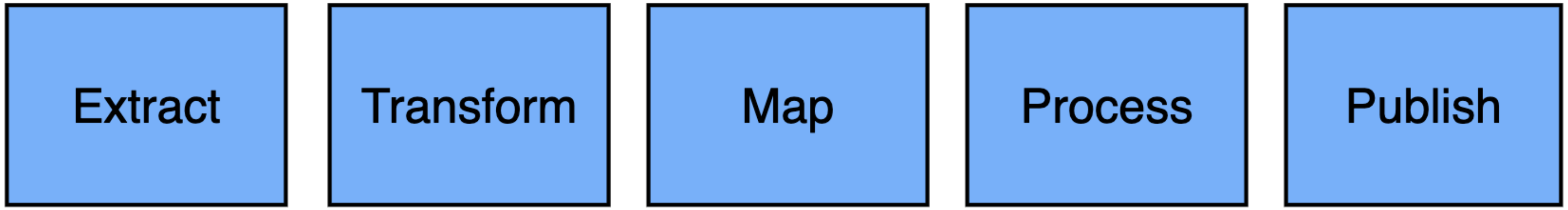
https://www.johnsnowlabs.com/clinical-nlp/

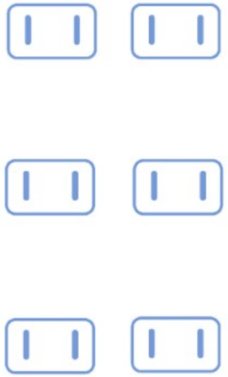
PROCESADO
DEL LENGUAJE NATURAL



https://openmed.life/

TRANSFORMACIÓN
AL MODELO OMOP-CDM





PUBLICACIÓN
DATA QUALITY DASHBOARD

DATA QUALITY ASSESSMENT

NJ

DataQualityDashboard Version: 2.8.0
Results generated at 2025-09-24 09:30:42 in 3 mins

	Verification				Validation				Total			
	Pass	Fail	Total	% Pass	Pass	Fail	Total	% Pass	Pass	Fail	Total	% Pass
Plausibility	515	7	522	99%	291	0	291	100%	806	7	813	99%
Conformance	912	3	915	100%	145	0	145	100%	1057	3	1060	100%
Completeness	477	7	484	99%	17	0	17	100%	494	7	501	99%
Total	1904	17	1921	99%	453	0	453	100%	2357	17	2374	**99%**

858 out of 2357 passed checks are Not Applicable, due to empty tables or fields.
0 out of 17 failed checks are SQL errors.
Corrected pass percentage for NA and Errors: 99% (1499/1516).

PUBLICACIÓN

ARES

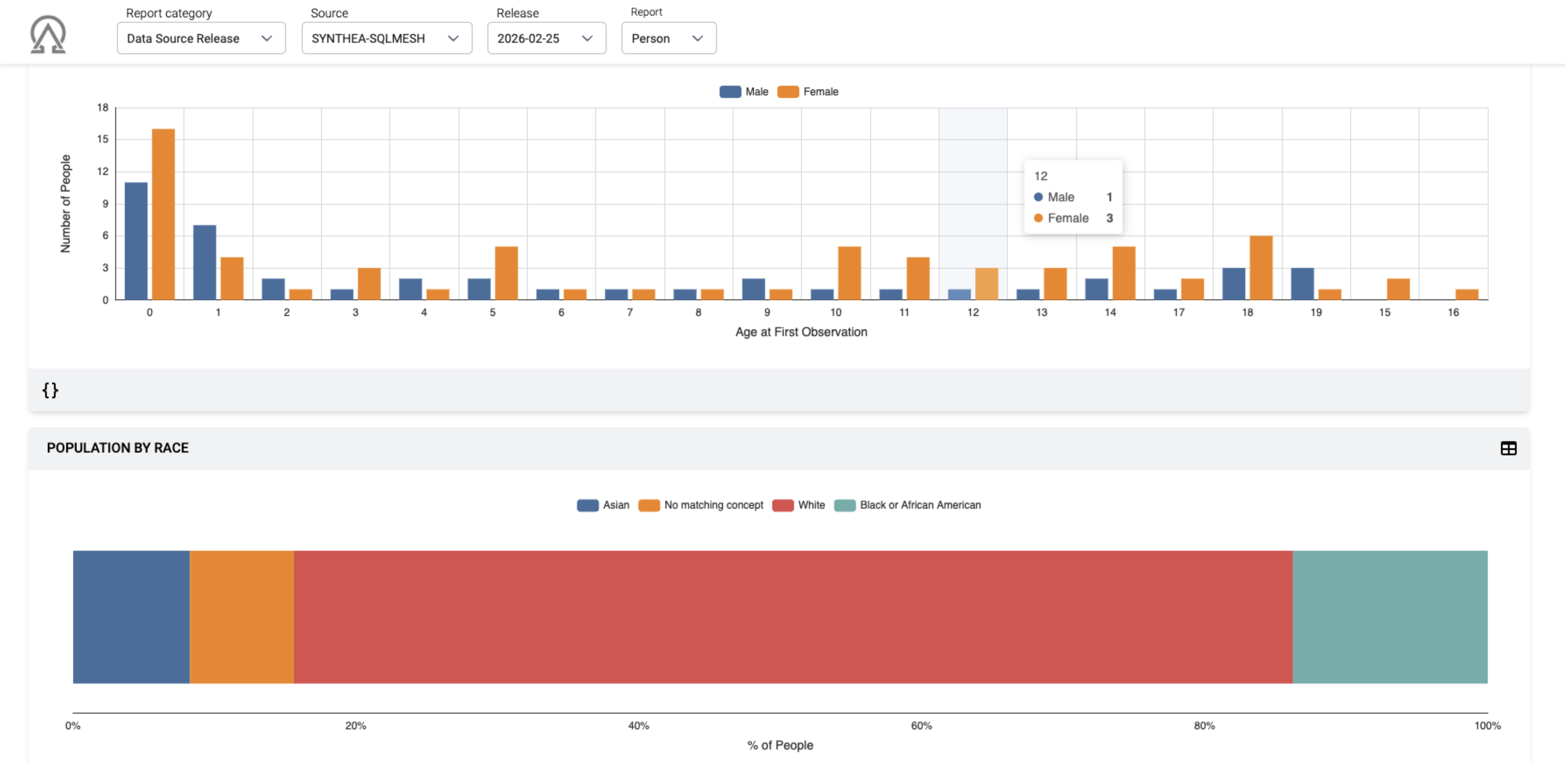
Report category: Data Source Release | Source: SYNTHEA-SQLMESH | Release: 2026-02-25 | Report: Data Quality

OVERVIEW | METADATA | RESULTS | PIVOT TABLE

	VERIFICATION				VALIDATION				TOTAL			
	PASS	FAIL	TOTAL	% PASS	PASS	FAIL	TOTAL	% PASS	PASS	FAIL	TOTAL	% PASS
PLAUSIBILITY	165	3	168	98.2%	207	0	207	100.0%	372	3	375	99.2%
CONFORMANCE	377	3	380	99.2%	55	0	55	100.0%	432	3	435	99.3%
COMPLETENESS	203	5	208	97.6%	8	0	8	100.0%	211	5	216	97.7%
TOTAL	745	11	756	98.5%	270	0	270	100.0%	1015	11	1026	98.9%

271 out of **1015** passed checks are Not Applicable, due to empty tables or fields
0 out of **11** failed checks are SQL errors
Corrected pass percentage for NA and Errors: **98.5%** (**744/755**)

PUBLICACIÓN
ARES



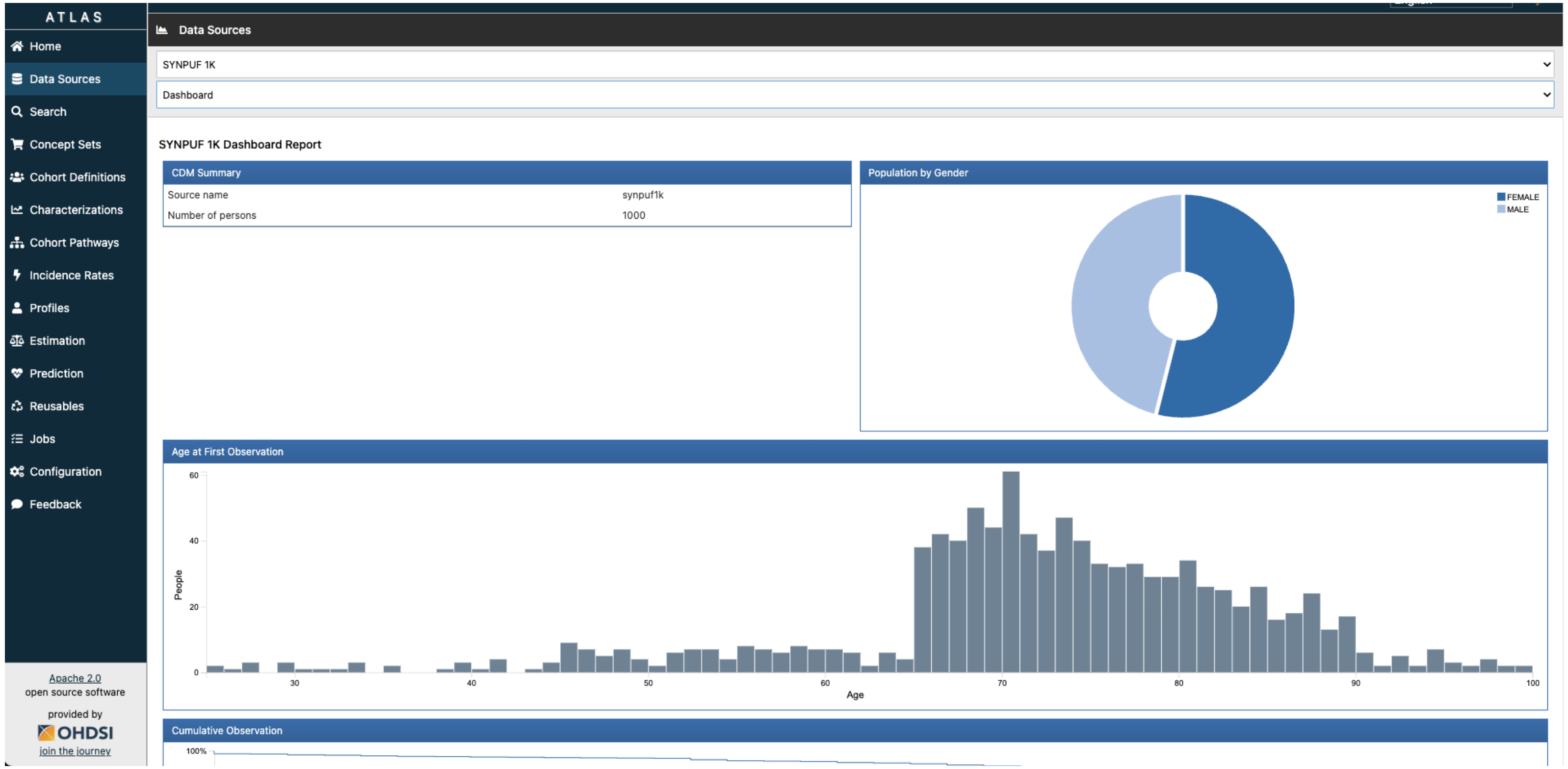
PUBLICACIÓN

ARES

```r
#Run data quality checks
DataQualityDashboard::executeDqChecks(
  connectionDetails = connectionDetails,
  cdmDatabaseSchema = cdmDatabaseSchema,
  resultsDatabaseSchema = resultsDatabaseSchema,
  vocabDatabaseSchema = vocabDatabaseSchema,
  cdmSourceName = cdmSourceName,
  numThreads = numThreads,
  outputFolder = datasourceReleaseOutputFolder,
  outputFile = "dq-result.json",
  verboseMode = T,
  writeToTable = T,
  cdmVersion = cdmVersion,
  tablesToExclude = tablesToExclude,
  sqlOnly = FALSE,
)
```

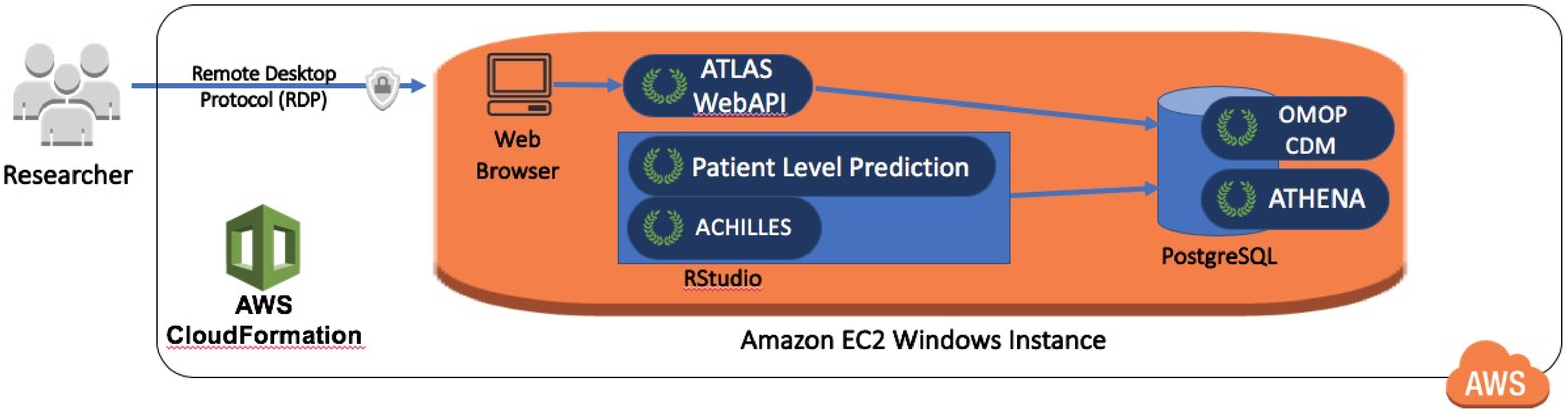
```r
Achilles::achilles(
  connectionDetails,
  cdmDatabaseSchema = cdmDatabaseSchema ,
  resultsDatabaseSchema= resultsDatabaseSchema,
  vocabDatabaseSchema = vocabDatabaseSchema,
  scratchDatabaseSchema = resultsDatabaseSchema,
  numThreads = 1,
  cdmVersion = cdmVersion,
  createIndices = F,
  createTable = T,
  smallCellCount = 0,
  sqlOnly = FALSE,
  defaultAnalysesOnly = TRUE,
  dropScratchTables = FALSE
)
```

PUBLICACIÓN
ATLAS



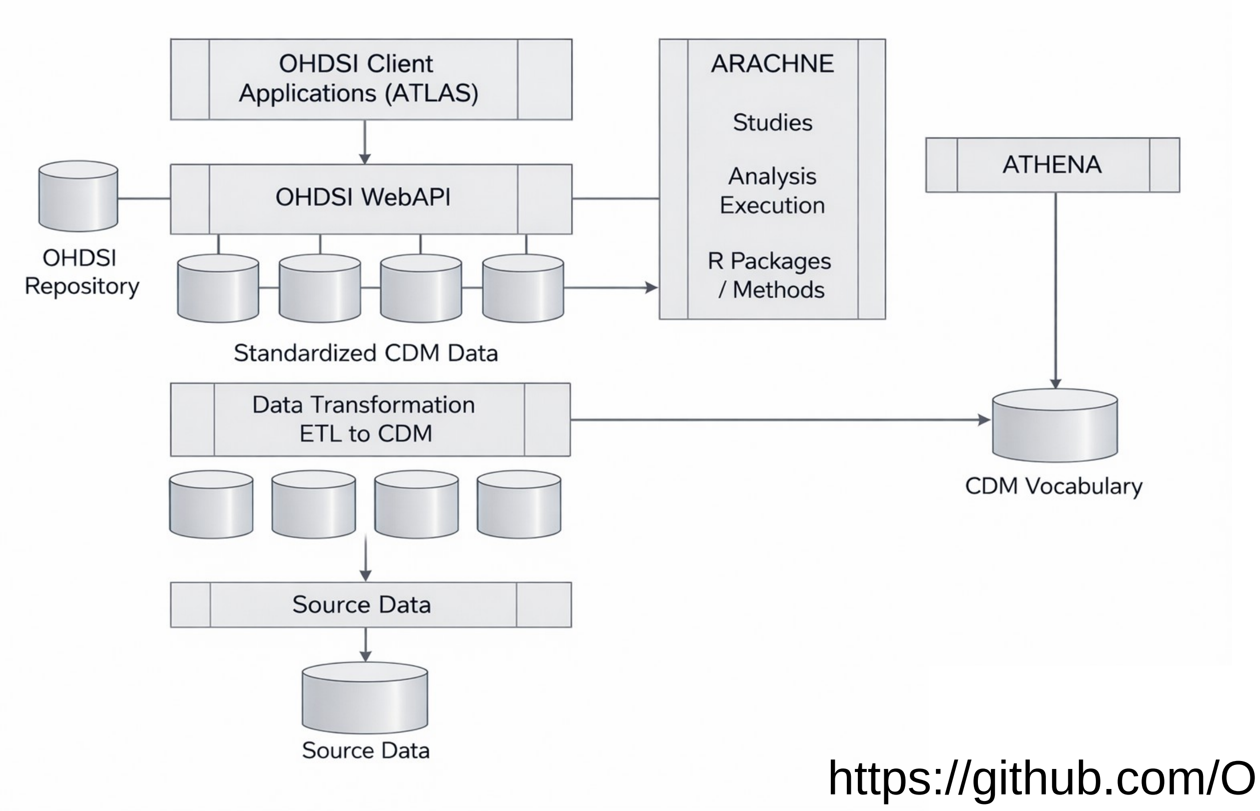
https://atlas-demo.ohdsi.org/

PUBLICACIÓN
OHDSI-in-a-BOX



https://github.com/OHDSI/OHDSI-in-a-Box

PUBLICACIÓN
OHDSI-in-a-BOX



https://github.com/OHDSI/Broadsea

Cohortes

Caracterización y predicción

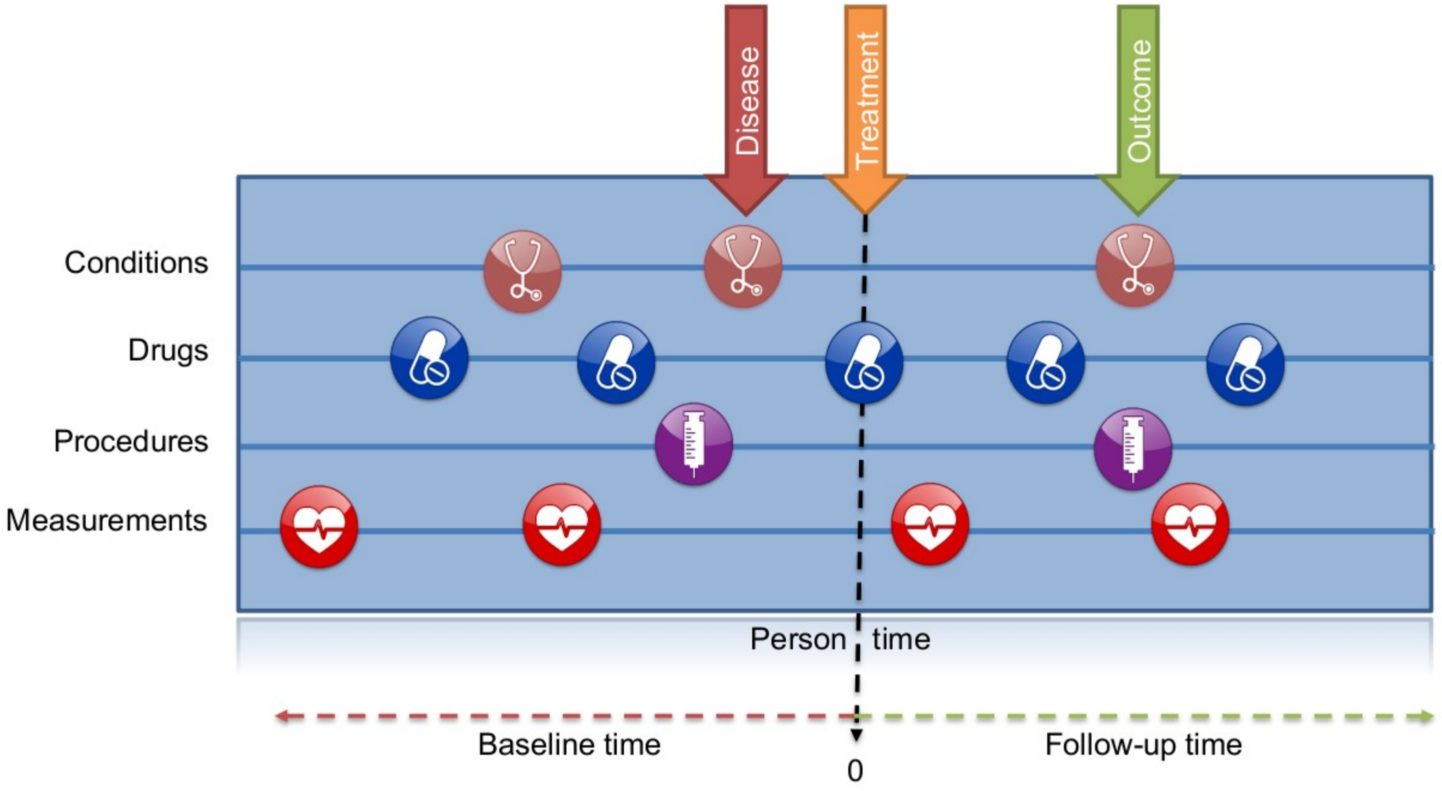
COHORTES
DEFINICIÓN

Un **cohorte** es un conjunto de personas que cumplen uno o más criterios de inclusión (un fenotipo) durante un período de tiempo.

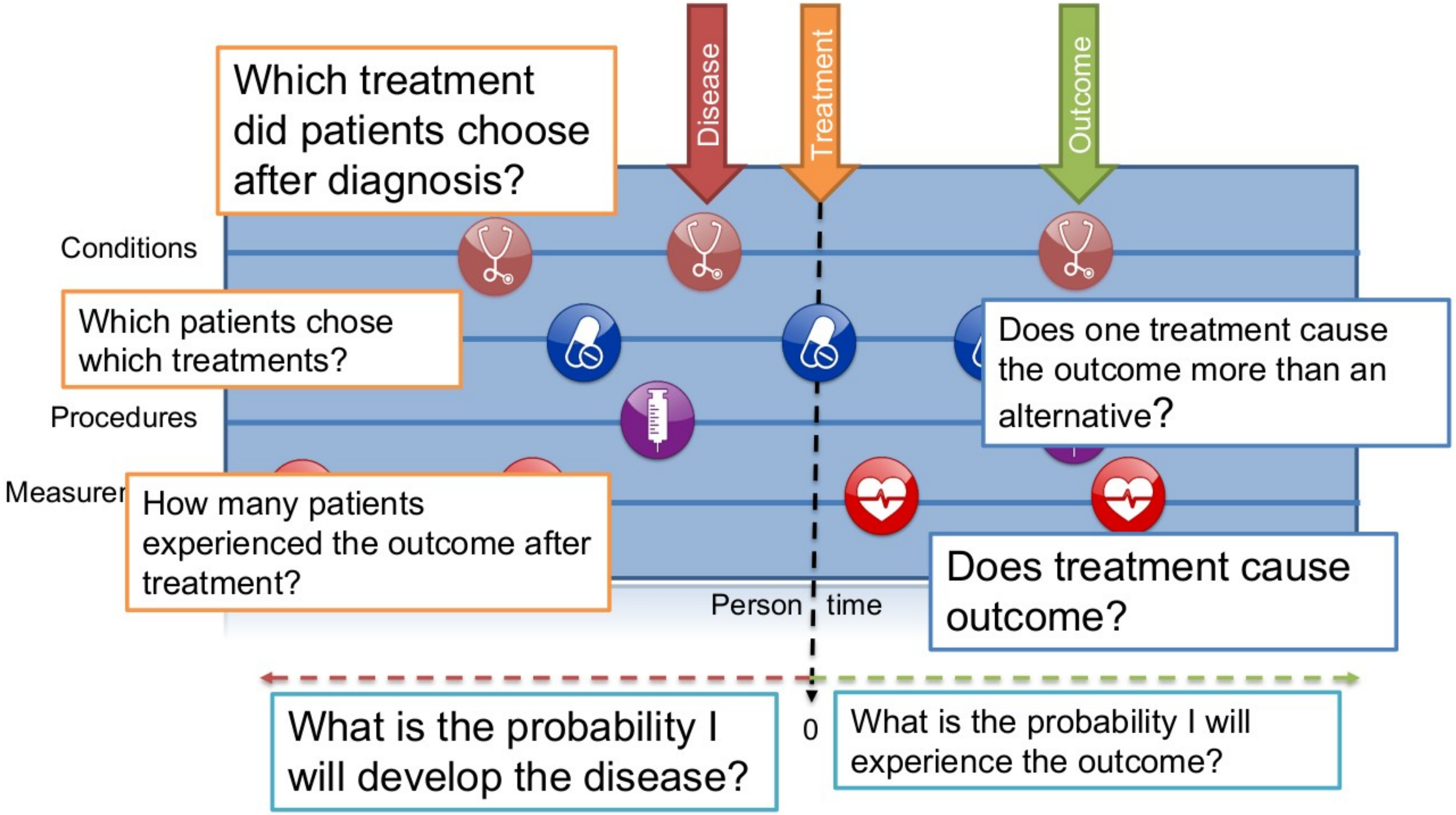
Un **fenotipo** es una especificación de un estado observable, potencialmente cambiante, de un organismo (a diferencia del genotipo, derivado de la composición genética). Muchas veces viene definido como un **concept set**

En nuestro caso serán las características de un paciente inferidas a partir de datos de registros médicos electrónicos (EHR, por sus siglas en inglés).

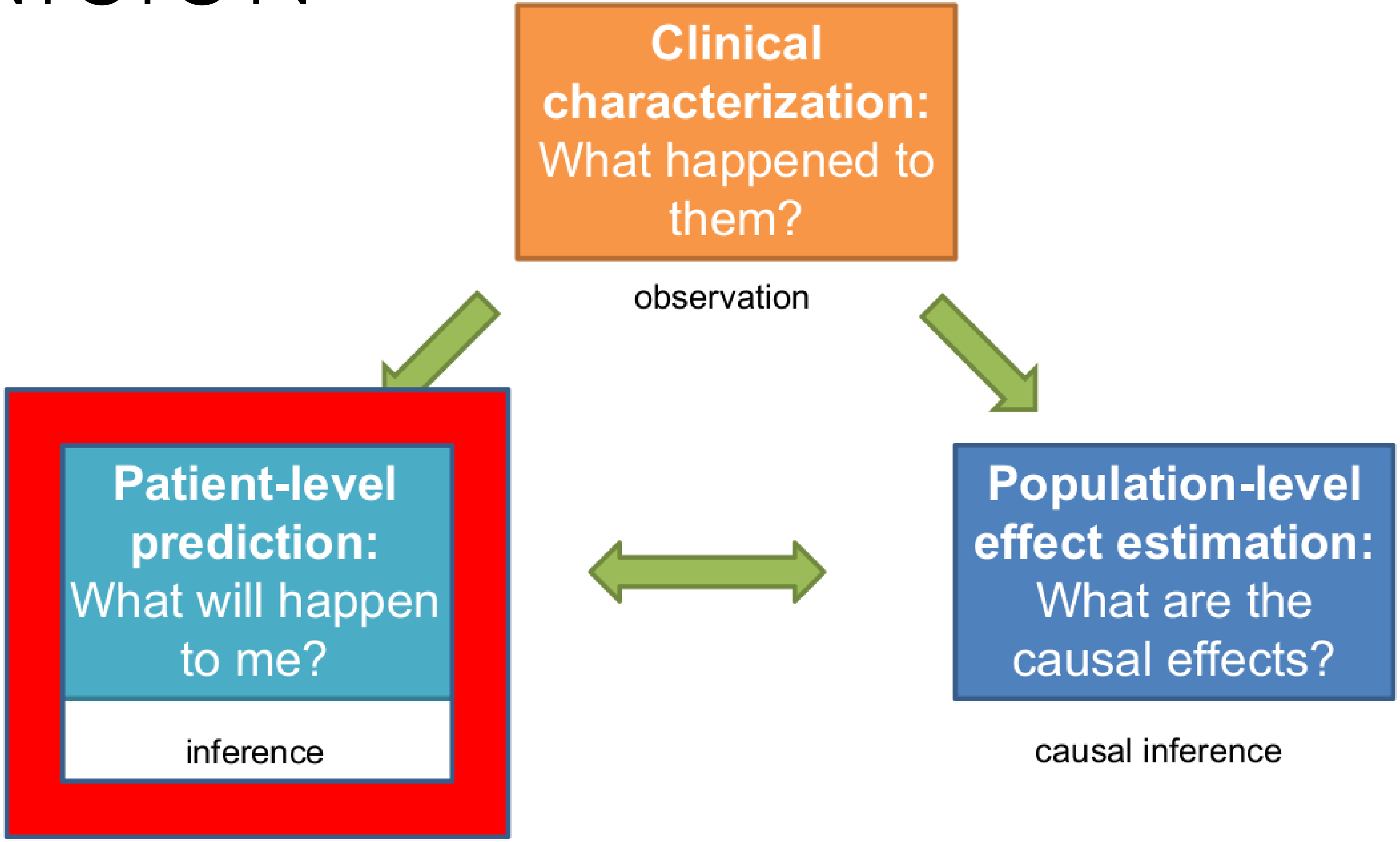
COHORTES
DEFINICIÓN



COHORTES
DEFINICIÓN



COHORTES
DEFINICIÓN



COHORTES
DEFINICIÓN

OBJECTIVE
To investigate the associations between direct oral anticoagulants (DOACs) and risks of bleeding, ischaemic stroke, venous thromboembolism, and all cause mortality compared with warfarin.

MAIN OUTCOME MEASURES
Major bleeding leading to hospital admission or death. Specific sites of bleeding and all cause mortality were also studied.

Target cohort:
- Person
- cohort start date
- cohort end date

What is the target (T) cohort?
People on Warfarin

Comparator cohort:
- Person
- cohort start date
- cohort end date

What are the comparator cohort?
Directly Acting Oral Anticoagulants (DOACs)
(dabigatran, rivaroxaban, apixaban - each is a cohort)

Outcome cohort:
- Person
- cohort start date
- cohort end date

Ischemic stroke
Venous thromboembolism
Hematuria
Haemoptysis
Lower GI Bleed
Death

COHORTES

DEFINICIÓN

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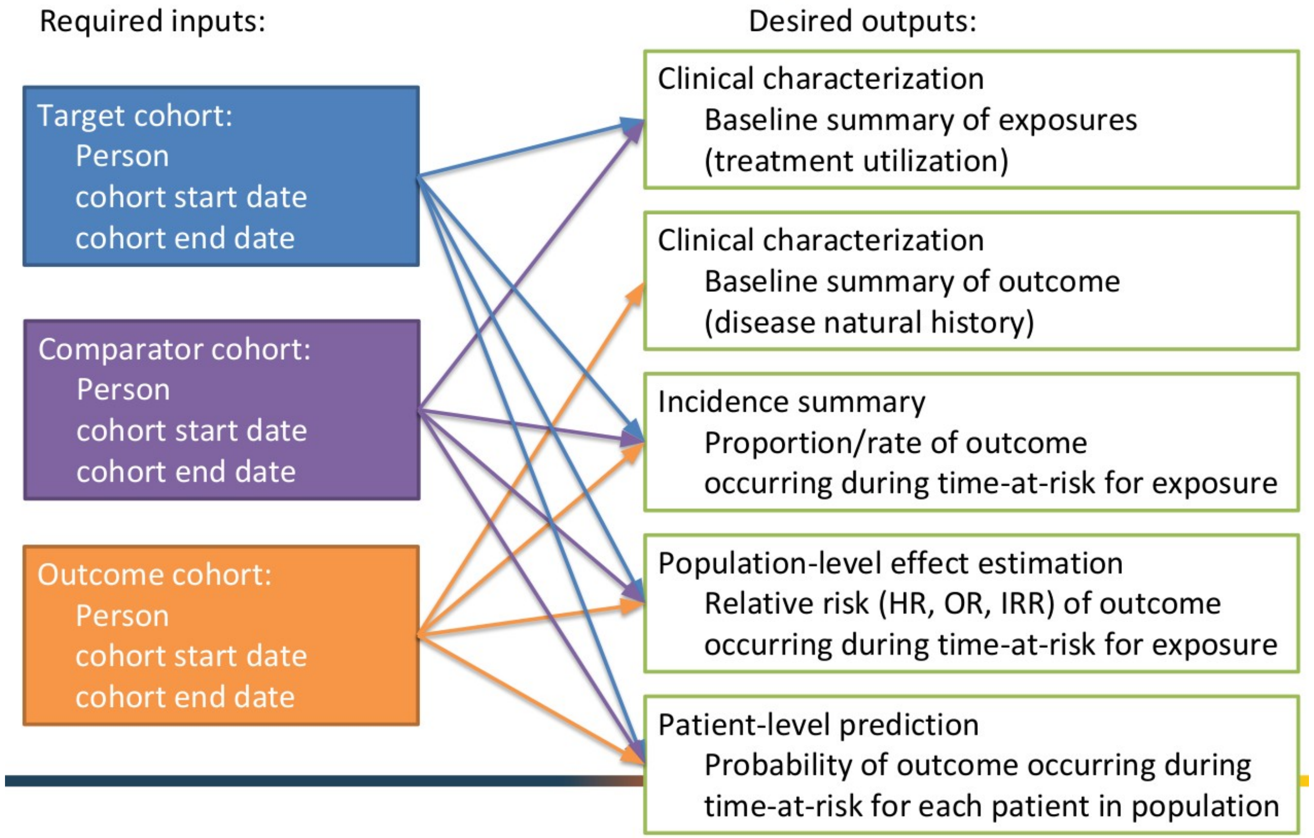
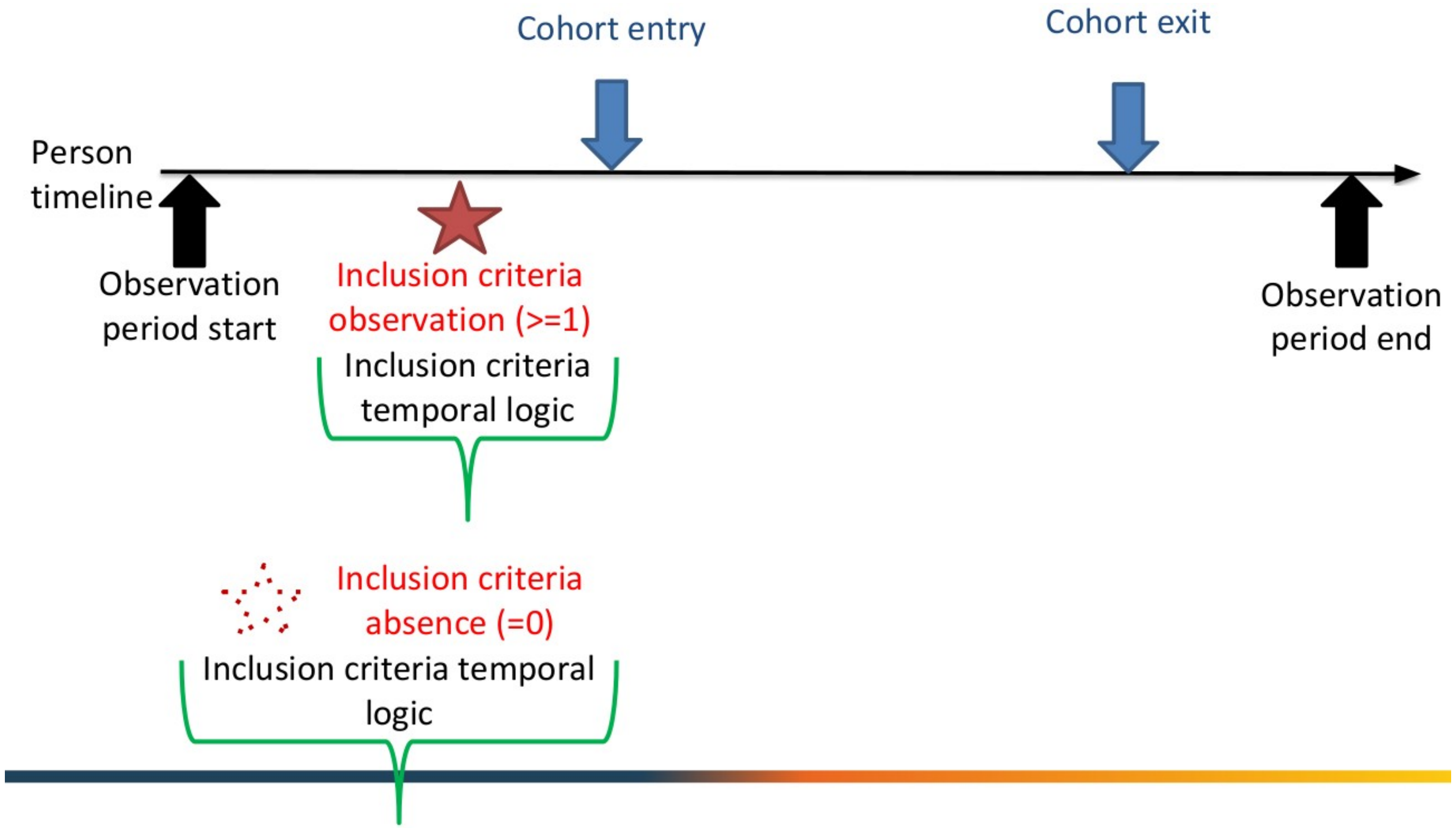
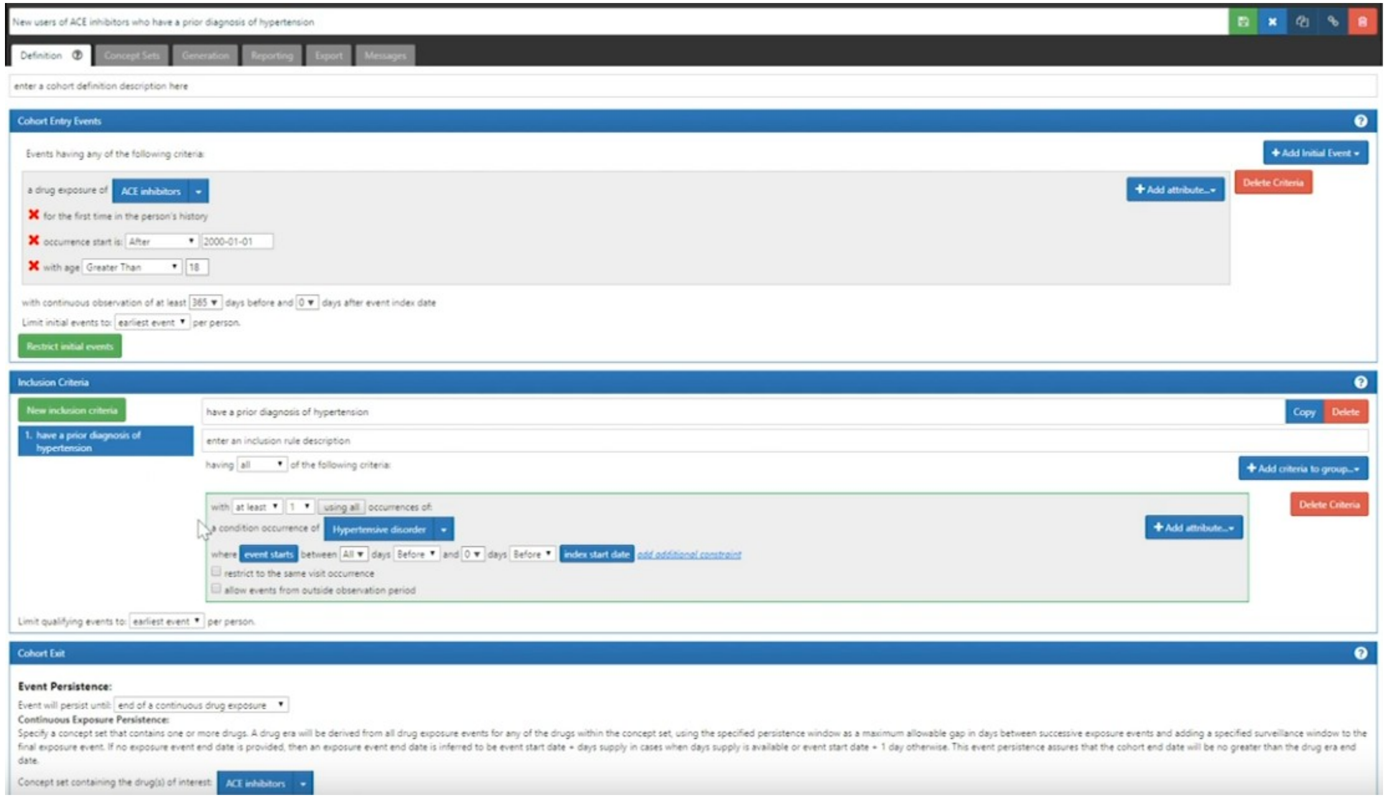
Major bleeding leading to hospital admission or death. Specific sites of bleeding and all cause mortality were also studied.



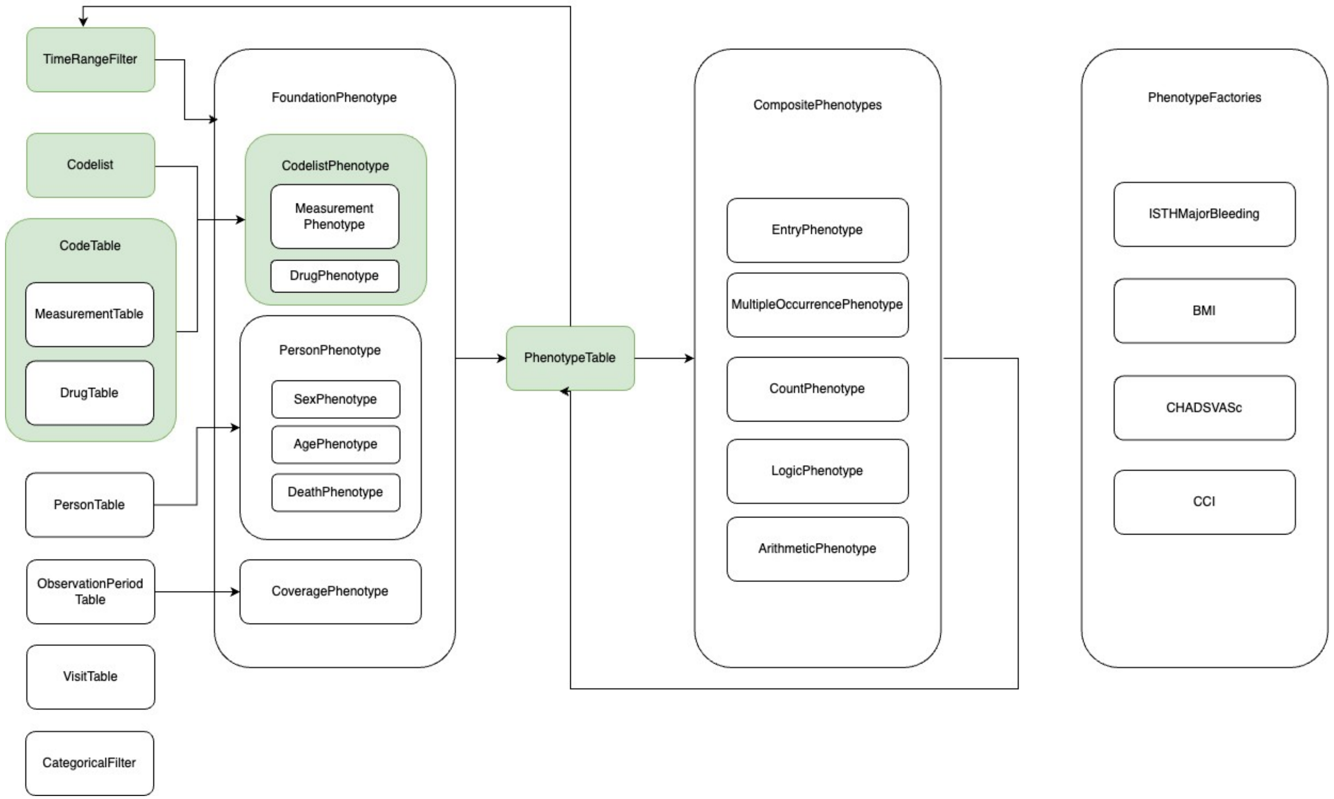
COHORTES
DEFINICIÓN



COHORTES
DEFINICIÓN

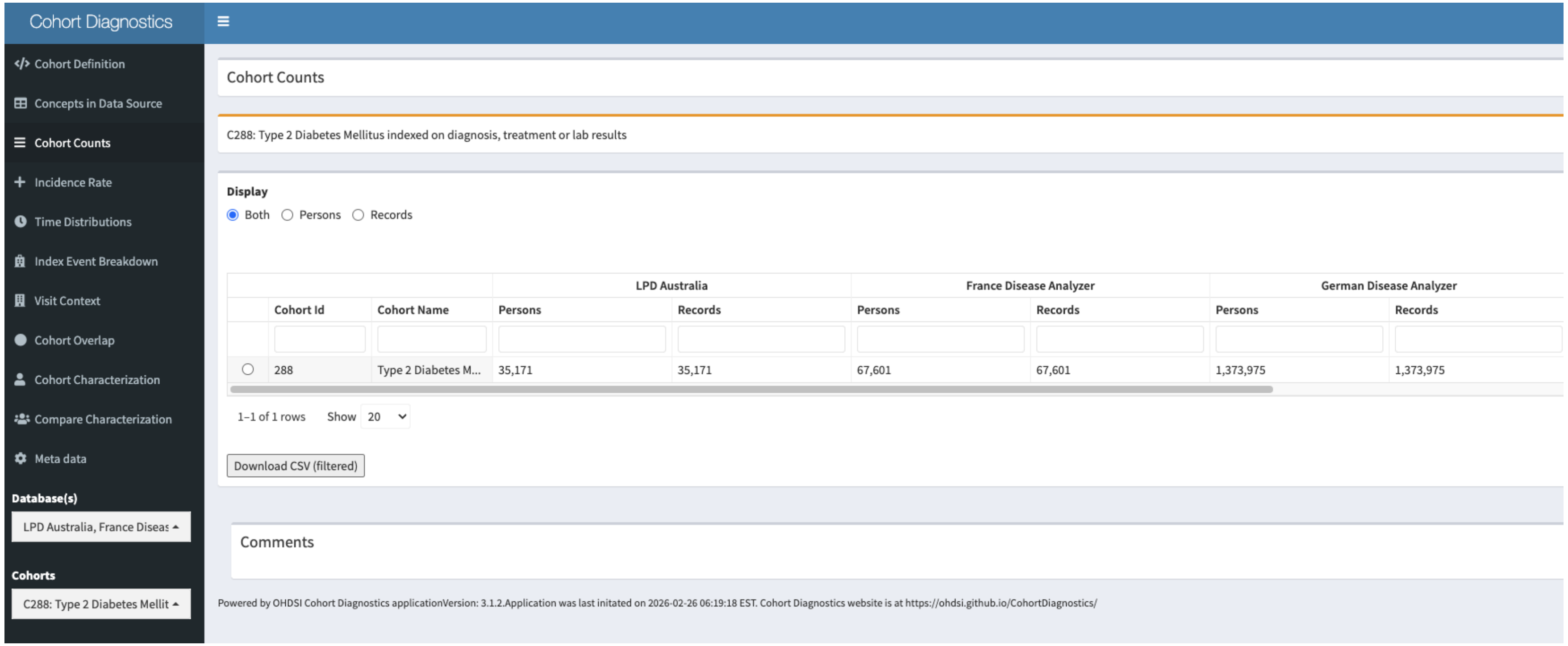


COHORTES
DEFINICIÓN

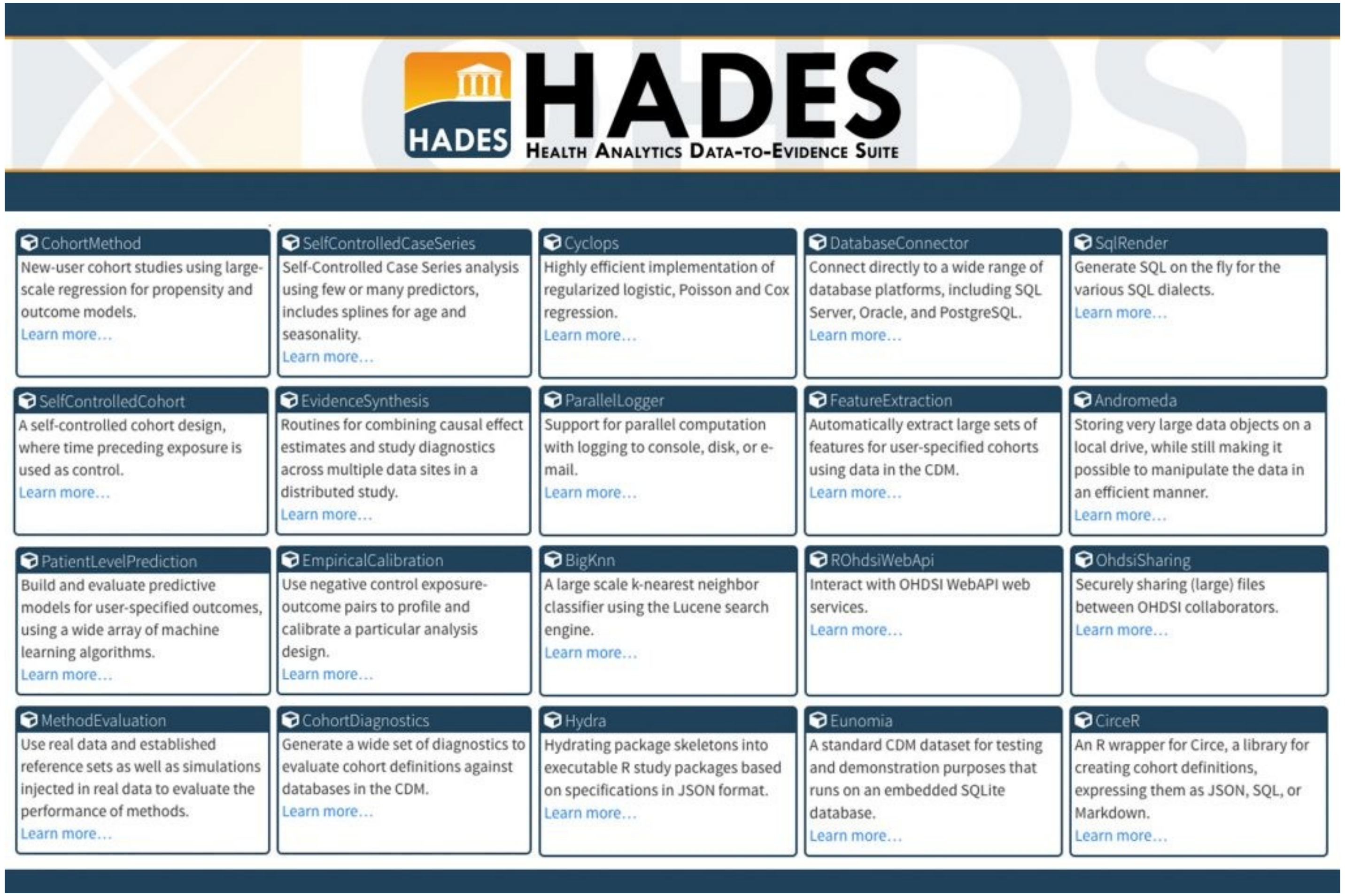


COHORTES

DEFINICIÓN



COHORTES
DEFINICIÓN



Preguntas?

GRACIAS POR VUESTRA ATENCIÓN